

Microclimates and Mosquito-borne Disease

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Background

Aedes aegypti is a species of urban, container-dwelling mosquitoes that transmit viruses that cause several diseases, including dengue, Zika, chikungunya, and yellow fever. *Ae. aegypti* population dynamics are strongly influenced by climate. Temperature, in particular, has a significant impact on mosquito life history traits. As an example, Mordecai et al. analyzed data from numerous studies and fit thermal performance curves to temperature-dependent trait data (Figure 1; [1]). Other researchers have used models to explore how temperature impacts *Ae. aegypti* population dynamics and dengue dynamics [2, 3, 4], emphasizing the importance of temperature for the potential for outbreaks.

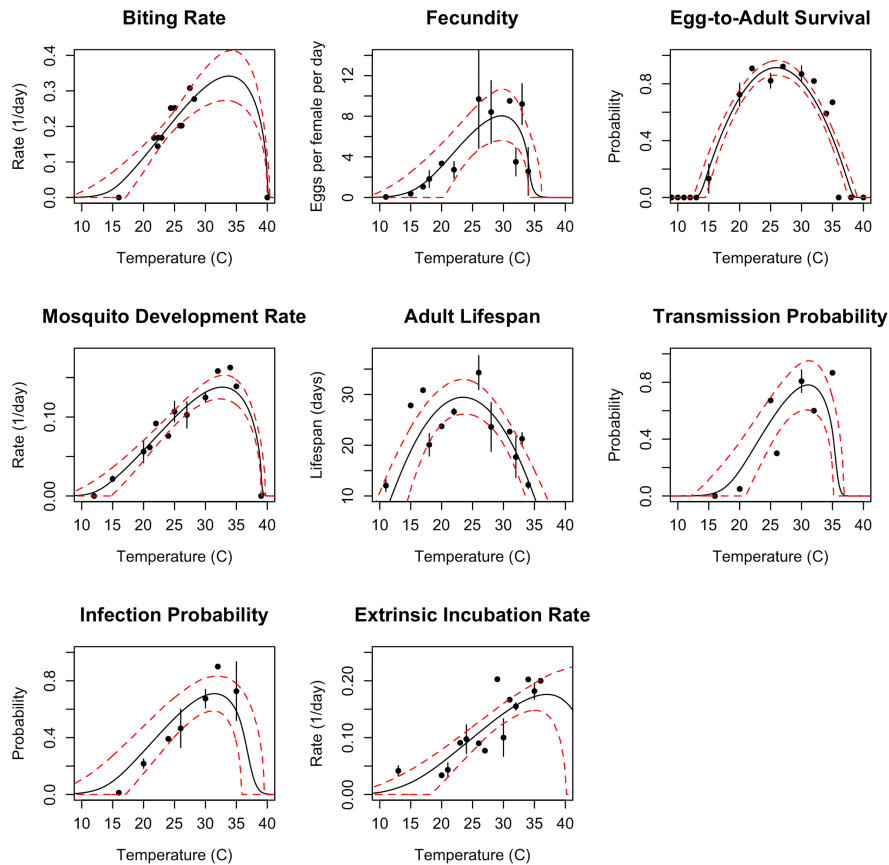


Figure 1: Some relationships between *Ae. aegypti* life history traits and temperature. From [1].

Given the propensity of *Ae. aegypti* to live in urban areas, population dynamics of the species may be strongly influenced by microclimates and Urban Heat Island (UHI) effects [5, 6]. UHIs are formed in cities by the presence of different types of groundcover and landscaping. For instance, concrete and asphalt often lead

Model

The diagram illustrates the transmission cycle of dengue fever, involving two populations: Humans and Vectors.

Human Population:

- Susceptible Humans** → **Infected Humans** → **Infectious Humans** → **Recovered Humans**

Vector Population:

- Juvenile Vectors** → **Susceptible Vectors** → **Infected Vectors** → **Infectious Vectors**

Transmission Pathways:

- Vertical Transmission (Humans):** A solid black line connects **Recovered Humans** back to **Susceptible Humans**, indicating immunity or return to susceptibility.
- Human-to-Vector Transmission:** A dashed purple arrow points from **Infectious Humans** to **Susceptible Vectors**, representing the bite of an infected human.
- Vector-to-Human Transmission:** A dashed blue arrow points from **Infectious Vectors** to **Susceptible Humans**, representing the bite of an infected vector.

Additional Vector Dynamics:

- Juvenile Vectors** have a black arrow pointing down, likely representing mortality.
- Susceptible Vectors** have a red arrow pointing down, likely representing mortality.
- Infected Vectors** have a red arrow pointing down, likely representing mortality.
- Infectious Vectors** have a red arrow pointing down, likely representing mortality.

Movement in the model will be incorporated implicitly. That is, we will not directly move people between populations but instead assume that humans have a home population but may spend some amount of time in other populations. Movement between two populations is determined by the gravity model, which calculates the rate of movement based on the population sizes of each population and the distance between them. There are two major assumptions behind the gravity model: (1) People are more attracted to larger populations than smaller populations, and (2) People are more likely to visit populations that are closer than they are to visit those that are farther away.

Project Goals

There are two primary questions that will be addressed in this project

1. How do microclimates and Urban Heat Islands impact the ability of dengue to spread in a city with interconnected subpopulations?
2. How do the timing and location of introductions of dengue impact outbreak size and spread throughout the city?

The answers to these questions are nuanced and depend on a number of model assumptions, parameterizations, and more! For this project, we will explore these two general questions. Below are some more specific questions to consider as you navigate the project. It is not expected that you will answer all of these, so feel free to choose the ones that are the most interesting!

General Questions

1. How does the location of introduction contribute to the spread of dengue in the area? How does the human population size (and the ratio of mosquitoes to humans) impact spread throughout the region? Introduce an initial dengue case in humans in each location separately and study how the outbreak spreads.
2. How does the timing of the first case in the year contribute to outbreak dynamics? Introduce dengue in humans at different times of the year in one population and observe how the outbreak spreads.

Changes in climate

The above questions can be explored in different climate scenarios. What happens if average temperatures increase? What happens if temperatures experience more extremes? In the model, the temperature T on day t is controlled by a cosine function:

$$T(t) = T_M - T_A \cos\left(\frac{2\pi}{365}(t + \tau)\right),$$

where T_M is the mean yearly temperature, T_A controls the amplitude of the temperature (the max and min temperatures will be $T_M \pm T_A$), and τ controls the timing of temperatures (the minimum temperature occurs when $t = \frac{2\pi}{365}\tau$).

1. How does changing the baseline mean temperature, T_M , impact dynamics?
2. How does changing the baseline amplitude of temperature, T_A , impact dynamics?
3. What other temperature scenarios can you consider? How would you incorporate these in the model? Try it!

Control Policy

Models are useful for testing different types of control policies. In an area with heterogeneous mosquito populations and disease risk, it may be advantageous to consider policies that prioritize control in certain populations. When using models to explore control policies, we can incorporate control in different ways. Here, we will think about how control impacts different parameters in the model.

1. Community education programs can influence how people protect themselves. We can explore this control measure by changing the probabilities of contract and transmission. In the model, these are included in the parameters β_{HV} and β_{VH} .

2. Community education programs can also lead to people removing potential breeding habitats. This might be controlled in the model with the parameter D_v , which influences the juvenile mosquito population carrying capacity. This could also be reflected by reducing the mosquito reproduction rate, α_v .
3. What are the impacts of applying pesticides? To explore this, consider modifying the adult mosquito mortality rate function $\mu_V(T)$. (Hint: one way to do this is to add a multiplier to the functions controlling lifespan and the adult mortality rate)
4. How can we prioritize locations for control in the model? What changes need to be made to the model to do this?
5. Mosquito control and community education can be expensive and time consuming. Can we use this model to determine the times of the season when control may have the biggest impact?
6. Are there other control strategies we can consider in the model?

Sensitivity to assumptions

We have made a number of assumptions in this model, and these assumptions have a strong influence on our results. Below are some questions to think about that would allow us to test the sensitivity of the model to these assumptions.

1. How does the choice of the gravity model to determine movement influence our results? What are some other ways movement could be considered here?
2. Temperature is modeled with a smooth curve. What impacts might this have? What types of climate phenomena may we miss by using this model? What are some other options for including temperature?
3. What are other environmental factors that may be important to include in the model? How can we include them?
4. Are there any other choices we made or assumptions that we should explore in further detail?

References

- [1] Mordecai EA, Cohen JM, Evans MV, Gudapati P, Johnson LR, Lipi CA, et al. Detecting the impact of temperature on transmission of Zika, dengue, and chikungunya using mechanistic models. *PLOS Neglected Tropical Diseases*. 2017 Apr;11(4):e0005568. Publisher: Public Library of Science. Available from: <https://journals.plos.org/plosntds/article?id=10.1371/journal.pntd.0005568>.
- [2] Robert MA, Christofferson RC, Weber PD, Wearing HJ. Temperature impacts on dengue emergence in the United States: Investigating the role of seasonality and climate change. *Epidemics*. 2019 Sep;28:100344. Available from: <http://www.sciencedirect.com/science/article/pii/S1755436518301609>.
- [3] Robertson SL, de Jesús Crespo R, Beck E, Beuerle L, Martin P, Stambaugh R, et al. Modeling the effects of vector species composition and temperature on the risk of dengue virus. *Bulletin of Mathematical Biology*. 2026;88(6):93.
- [4] Beck E, Beuerle L, Martin P, Stambaugh R, de Jesús Crespo R, Robert MA, et al. Modeling the Effects of Temperature and Resource Quality on the Outcome of Competition Between *Aedes aegypti* and *Aedes albopictus* and the Resulting Risk of Vector-Borne Disease: E. Beck et al. *Bulletin of Mathematical Biology*. 2025;87(10):138.
- [5] de Jesús Crespo R, Pavlakakis A, Breaux J, Riegel C. Discarded vehicle tires and their association with mosquito vector abundance across socioenvironmental gradients in New Orleans, LA. *Journal of Medical Entomology*. 2024;61(5):1240-50.

- [6] Sangermano F, Asar ML, Visintin AM, Benítez E, Ludueña-Almeida FF, Estallo EL. Temporal and spatial dynamics of urban heat in the City of Córdoba, Argentina. The Journal of Climate Change and Health. 2025;26:100599.

Model Equations

The equations for the human infections dynamics are:

$$\begin{aligned} S_h(t+1, i) &= S_h(t, i) - \lambda_{vh}(t)S_h(t, i) \\ E_h(t+1, i) &= E_h(t, i) + \lambda_{vh}(t)S_h(t, i) - \sigma_h E_h(t, i) \\ I_h(t+1, i) &= I_h(t, i) + \sigma_h E_h(t, i) - \gamma_h I_h(t, i) \\ R_h(t+1, i) &= R_h(t, i) + \gamma_h I_h(t, i) \end{aligned}$$

The mosquito dynamics are more complicated because we consider mosquito mortality, which means we have to consider the probability of mosquito survival in the movement of mosquitoes between stages. The equations for mosquitoes are:

$$\begin{aligned} J_v(t+1, i) &= J_v(t, i) + \alpha_v N_v(t, i) - (1 - f(J_v(t, i))\nu_v(t)J_v(t, i) - f(J_v(t, i))J_v(t, i) \\ S_v(t+1, i) &= S_v(t, i) - (1 - \mu_v(t))\lambda_{hv}(t)S_v(t, i) - \mu_v(t)S_v(t, i) \\ E_v(t+1, i) &= E_v(t, i) + (1 - \mu_v(t))\lambda_{hv}(t)S_v(t, i) - \sigma_v(t)E_v(t, i) - \mu_v(t)E_v(t, i) \\ I_v(t+1, i) &= I_v(t, i) + (1 - \mu_v(t))\sigma_v(t)E_v(t, i) - \mu_v(t)I_v(t, i) \end{aligned}$$